Novel strategies for the point decomposition problem: a desk evaluation against the lead atlas

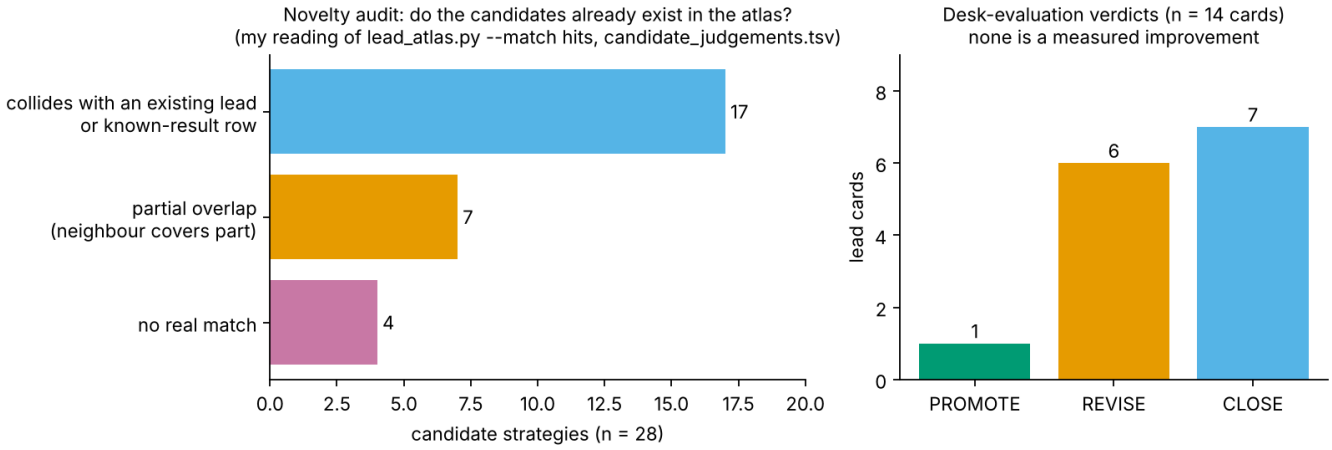
Date. 2026-10-31. **Question (owner).** "Can we evaluate any novel strategies here for PDP?" **Status.** Desk evaluation. Nothing in this directory is a measured improvement, and none is claimed (success criterion: a verified, measured improvement over the current index-calculus baseline; the gap to rho is distance remaining, never a gate). **Tools.** No build of the crate was possible: the container's disk allowance was spent, so every number below is either read from a committed record, derived from one, or taken from a primary source.

1. Answer in six lines

1. **Nothing here is new to the repository.** Of 28 candidate strategies I could name (the three I suggested earlier, ideas from the PR #2423 survey, and further brainstorm), 17 collide with an existing atlas lead or known-result row, 7 partly overlap one, and 4 have no real match (§2). The four with no match are the ones I rate least promising (§2.2).
2. **The atlas does not contain the merged PR #2423 work** (LEAD-19, the 19-lead survey): a search for `lead19`, `projected base` and `pdp_literature_leads` in `knowledge/leads/ecdlp/` on `main` (2026-10-31) finds nothing. A draft lead row for it is in §6.
3. **Why replacing summation polynomials cannot move today's baseline.** The ladder rows are not solved with summation polynomials at all. They use a pair table and a walk, and on arm B at the published rung the walk is 61–82 % of S and the matrix is at most 0.10 % (§3, measured). An algebraic-oracle lever (solving degree, hybrid guessing, SAT, a tensor encoding) cannot touch those terms. On the committed algebraic-oracle rows the oracle's share of S is exactly zero when native operations are unpriced, so its ceiling is 1.00× (cards/e1.md appendix, derived).
4. **Of 14 lead cards (13 atlas leads plus my LEAD-15): 1 PROMOTE, 6 REVISE, 7 CLOSE** (§4). The one promotion is LEAD-15, the only candidate that changes an exponent on a family the repository can build, with a decisive first test that costs under one CPU-hour once an instance builder exists.
5. **The cheapest decisive item changes an existing baseline, not a lead.** The committed volcano study runs its IC arm on a raw (unprojected) factor base over curves with cofactor 4 and 12, which is the regime LEAD-19 showed wastes a fraction `1 - 1/h` of every folded table (§5.2). I verified the premise; the size of the effect is unmeasured.
6. **Corrections found** in the atlas, in my LEAD-15 dossier and in my own brief to the evaluators (§7).

2. Novelty audit

Method: each candidate was run through `python3 tools/lead_atlas.py --match` (153 leads, reviewed 2026-10-30; sparse clone of crypto-autoresearcher `main`). The matcher is lexical, so I read every top hit and recorded my own reading in candidate_judgements.tsv the raw hits are in atlas_match_log.tsv.



Novelty audit and verdicts

2.1 What the atlas already holds on the three directions I suggested earlier

My earlier answer to "other areas of PDP" proposed odd-characteristic covers, degree of regularity as a search objective, and hybrid guessing. All three are atlas leads: `LEAD-TRF-odd-prime-ghs-cover-floor` (open), `LEAD-ICE-linearization-excess-fb-prior` (open_unexamined) and `LEAD-ICE-hybrid-guess-determine-l7` (open_next_step). That answer was given before I had read the atlas; it was not new.

2.2 The four with no match

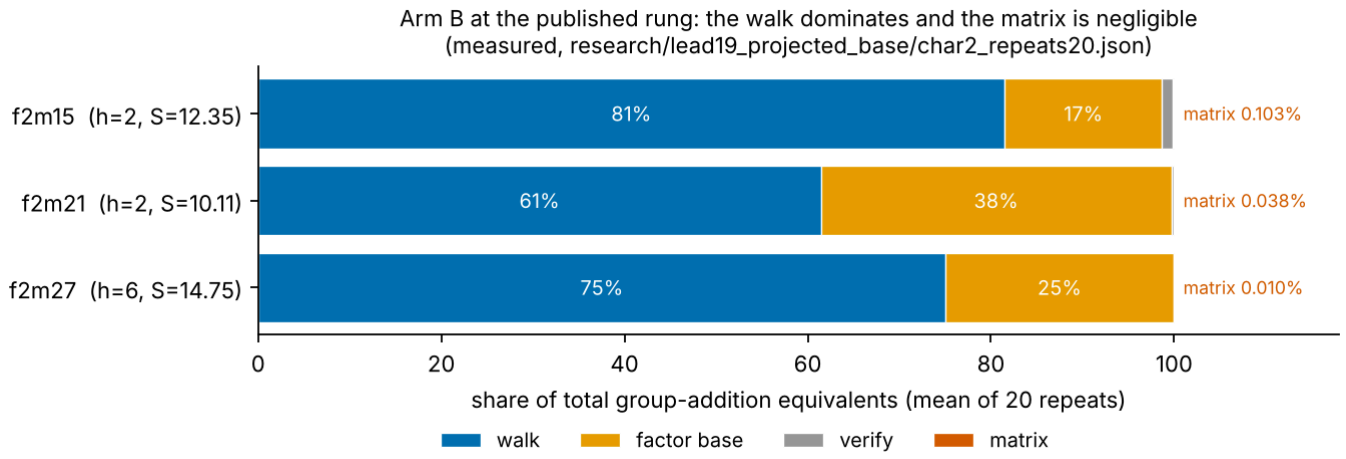
These are my first-principles notes, **not verified and not evaluated by an agent**:

- **Sieving over a lattice of x-coordinates instead of solving.** Closest lead: `LEAD-TRF-jv-sieve-fp6-weak-class` (a sieve inside one weak family). For a prime-degree binary extension there is no smoothness notion on `x` to sieve, so I see no object to sieve.
- **Relations on the quadratic twist.** A twist point gives a relation in the twist's group, which does not involve the target's discrete log; I see no use.
- **List decoding (Guruswami–Sudan) as a formulation.** There is no received word. "All roots in the subspace V" is the divisibility `h | L_V` of the linearised polynomial, which is the form `LEAD-ICP-shared-coefficient-membership` examines (verdict REVISE, expected null, §4).

- **Hankel / Berlekamp–Massey rank method.** I find no linear recurrence to exploit; the matrix methods that apply (Macaulay, bilinear XL) are already in the repository.

3. Where S goes

Recomputed from `research/lead19_projected_base/char2_repeats20.json` by phase_shares.py (output in phase_shares.json and phase_shares.md). Arm B at the published rung, mean of 20 repeats:



Phase shares

row	variant	S	factor base	walk	matrix	verify
f2m15 (h=2)	…_walk_projected	12.35	17.2 %	81.5 %	0.103 %	1.21 %
f2m21 (h=2)	…_walk_projected	10.11	38.3 %	61.4 %	0.038 %	0.26 %
f2m27 (h=6)	…_walk_projected_balanced	14.75	24.9 %	75.0 %	0.010 %	0.05 %

Consequences (the arithmetic is in cards/e1.md and cards/e3.md):

- The levers that remain on these rows are cost per candidate in the walk, the number of targets K and target sharing, and walk restarts. The yield per group addition is fixed (KN-FIND-007).
- On the three-summand rows the same record gives `S(m=3 walk) / S(m=2 walk)` = 1.62, 1.70, 2.53, 2.23, 3.90 on the raw base (ratio of means), so three-summand probes are not a free upgrade.
- An oracle-only gain has a ceiling of exactly 1.00× when native operations are unpriced, and the oracle's share exceeds 50 % only above about 3e-6 to 7e-5 GAE per word XOR (E1 appendix; derived).

4. Verdicts

Fourteen cards from five evaluators (Sonnet 5.5, desk only; cards are unedited in cards/). "Decisive test" is the evaluator's, with its frozen kill criterion.

lead	verdict	one-line reason	decisive test (cost)
LEAD-15 Diem–Thomé plane quartic on the cubic-GHS cover	**PROMOTE**	the one exponent candidate on a buildable family (Õ(q) vs Õ(q^{4/3}) for the hyperelliptic genus-3 baseline); model predicts 9–40× below the hyperelliptic incumbent at $q=2^{22}$	instance builder ($q \in \{257, 2053, 32771, 2^{22}\}$, 6 curves each), then the full harvest; kills K1–K4 on pair yield, cycle count, nullity, projected total; run < 1 CPU-h, builder 2–3 person-days
LEAD-TRF-odd-char-descent-smith-quartic	REVISE	dominated by LEAD-15; atlas text misstates the exponent; blocker stale	W3a Q1 exhaustive for $p \in \{7…23\}$; kill pass(E) < 10 % (hours)
LEAD-ICE-linearization-excess-fb-prior	REVISE	baseline subspace already is the top-e one; gain over it is 0; a stage diagnostic cannot move S	redraw the ℓ=7 cells (m=3, n=19, 21) with prefix and geometric bases, 8 UNSAT draws each; kill ≤ 3/16 (1–2 h, 5–8 GB)
LEAD-ICP-shared-coefficient-membership	REVISE	expect a null: per-root tower grows 2.4–2.5 bits per unknown vs a 0.30 budget	ladder over t = 3, 4, 5; kill ≥ 1.5 bits per unknown
LEAD-ICE-hybrid-guess-determine-l7	REVISE (low)	arithmetic reproduced; E8 can close the lead but never promote it	extend R1 past 3 variables, ℓ=8 on ≥ 2 UNSAT draws; kill any $k^* \geq 3$ (6–12 h, 7–12 GB)
LEAD-ICB-nonlinear-frob-stable-bases	REVISE (low)	zero effect on any current row; uncited prior art refuted the same idea	scaling-first diagnostic on $n \in \{19, 29, 37, 53\}$; kill R(n) ≥ 0.5 by n=37 (CPU-hours)
LEAD-STR-class-representative-ic-selection	REVISE	cofactor is an isogeny invariant; the 4.5 % S effect may be an artefact of the raw base	rerun the ℓ=2 volcano exports with `prime-projected-abscissa`; kill CI inside [0.97, 1.03] (≈ 2.4–7.7 CPU-h)
LEAD-ICE-decomposition-probability-audit	CLOSE	yield is size-only (KN-FIND-007); headroom ≈ 0.1 % at the Table B rows	none for S (batch BATCH-79a580 only retires an instrument)

LEAD-ICE-semaev-yield-model-coset-typing	CLOSE (typed half)	typing is +0.4 to +1.7 bits worse under the committed model; the trace-parity half is already realised by LEAD-19	new engineering test: `geomtrace` base in the algebraic arms; kill < 1.10× paired S across the pricing band
LEAD-ICP-xline-symmetry-special-j	CLOSE	automorphism fold is a measured constant (÷3 at j=0); rank can only drop unfavourably	none worth running (optional fibre-rank check, < 1 CPU-h)
LEAD-EXO-four-list-ktree-join	CLOSE (ladder)	only homomorphic filter is the cofactor class, already spent by LEAD-19; predicts S ≥ 1.00×	none on the ladder
LEAD-ICB-kic-golden-base-exact-decomp	CLOSE (baseline lever)	coverage cannot exceed the null; at most ≈ 2 % of S	none for the baseline
LEAD-EXO-hashed-factor-base-tensor-ecdl	CLOSE (watch item)	conditional on an unreviewed theorem; saving needs ≥ $2^{[2}17\}$ edges	none; the 84-case panel cannot discriminate
LEAD-TRF-odd-prime-ghs-cover-floor	CLOSE (improvement)	n=7 floor already known; for g=n Diem's exponent equals Gaudry's	none; census is formalisation hygiene

4.1 LEAD-15, the promotion, with the evaluator's corrections

- Diem's plane-quartic result is Õ(q) (heuristic in Diem 2006, ePrint 2005/119, a theorem in Diem, JTNB 2012); the hyperelliptic genus-3 bound is Õ(q^{4/3}). I checked these two exponents against the search result; I did not read the JTNB theorem statement myself.
- Corrections to my LEAD-15 dossier (E5): the provable variant costs at least 322q additions, not about 48q; "rho about 10^5× more work" is against relation collection only (Diem–Thomé's end-to-end is about 1.6e3); the 12 non-zeros per row is the weakest assumption (Diem–Thomé measured 352 at q = 2^{31}); and the baseline is stale: round 4 has an incumbent slope of 1.408 over 2^{16}–2^{22}.
- Predicted total at q = 2^{15} is 7.3 M + 0.53 M·w multiplications (derived): 13.6 M, 28 M, 60 M for w = 12, 40, 100 non-zeros per row, i.e. 0.12 to 0.55 of the hyperelliptic incumbent.
- **Family gap.** It needs `E/F_{q³}` with 4 | #E, so never prime order, and that family is not on Tables A–C. A promotion creates a new row, not an improvement to an existing one. Its baseline is the round-4 hyperelliptic genus-3 row on the *same instances*, which the first test arranges.
- In the tower family `F_{p⁶}/F_{p²}/F_p` Joux–Vitse's own exponents (p^{5/3}, p^{12/7}) beat Diem's p^2, so the quartic route wins only when q has no subfield.
- **A later in-repo measurement exists and the evaluator could not see it.** The shared session log of 2026-10-26 (entry "Hess cubic product-family audit", Conductor T-5446, PR #2651) transferred two q = 8 sources (subgroup primes 137 and 139) to smooth canonical plane quartics and ran Magma's line sieve end to end. The median full transfer cost 0.720 s and 0.870 s against a same-source rho of 0.0003125 s, a loss of 2,304× and 2,784×. That is a characteristic-2, q = 8 control far below the modelled crossover (q ≈ 2^8 to $2^{11.5}$), so it neither supports nor contradicts the promotion; it is the first measured quartic-transfer cost in the repository and should be read before the instance builder is written. I read the log entry, not the audit's `RESULTS.md`.

5. Two findings that do not depend on any lead

5.1 No priced oracle-side scenario crosses rho at ECC2K-130

From the committed cost model (cards/e1.md): for m=4, one closure degree is worth 2.7 bits at one guessed variable (+10.32 ⟶ +7.62 over rho) and 12.6 bits at two (+45.71 ⟶ +33.10); the best case D=3 reads +5.34 and +19.84. None of these priced rows is below rho. Only a *free* oracle would be: the model's free-oracle floor at m=4 is 2^56.46, 4.35 bits below rho, because targets × oracle is 26 % of the D=4, w=1 cost and linear algebra is the rest. Note the correction in §7: the `+7.62` is the D=4, w=1 row, not a floor.

5.2 The volcano IC arm runs the raw base on cofactor-4 curves

Verified by me from the committed specs: `research/isogeny-volcano-cost-v2_20261017/sessions/*/spec/spec.json` set `"factor_base": "prime-abscissa:size=32"`, and the first curve there has `group_order` 10967284 and `r` 2741821, so h = 4. `prime-projected-abscissa` is a registered plug-in (`src/cryptanalysis/ecbench/methods.rs:2159`). The evaluator (E4) reports that all 14 ℓ=2 classes have h ∈ {4, 12} (not re-checked by me) and that the session mean is 11.5 relations from 85,707 targets, which is the dead-weight regime of LEAD-19. So the volcano study's 4.5 % S effect between curve models was measured on a base that wastes about 75 % (h = 4) of each table. Whether the effect survives a projected base is the decisive test of `LEAD-STR-class-representative-ic-selection`. If the LEAD-19 mechanism carries over to the prime abscissa base the baseline of those rows is also improvable; **this is unmeasured**.

6. Atlas follow-up

`LEAD-19` is merged but absent from the atlas, and so are the survey's LEAD-15 and the findings above. A draft row for the cofactor-projected ladder base is in atlas_drafts/LEAD-ICB-cofactor-projected-ladder-base.yaml. The existing `LEAD-ICB-cofactor-torsion-quotients` (closed_obstructed) is **not contradicted**: its obstruction is about ECC2K-130 and standardized prime-degree curves, where the cofactor carries two bits and gcd(4, l) = 1; the ladder rows are random binary curves with h = 2…8. Its `revisit_when` ("a curve with a large smooth cofactor") is the neighbouring regime.

7. Corrections

where	correction	source
my brief to the evaluators	"+7.62 floor" is the D=4, w=1 row; the free-oracle floor at m=4 is lower (−4.35 bits)	E1
atlas `LEAD-TRF-odd-char-descent-smith-quartic`	title, mechanism and `discriminates` attribute q^{4/3} to the plane quartic; that is the hyperelliptic bound (quartic: Õ(q)). `blocked_by` is stale: Smith Props 2 and 4 are native	E5, exponent re-checked by me

atlas `LEAD-ICE-semaev-yield-model-coset-typing`	`EV-SEMBIN-36f27d` is cited as "coset typing near finite-k cap"; the record says admission gates, direction neutral	E1
atlas `LEAD-EXO-four-list-ktree-join`	`best_measured` repeats the wording `KN-FIND-ffe1df` withdrew	E3
atlas `LEAD-ICP-xline-symmetry-special-j`	evidence omits the measured ×3 fold at j=0; `H-PFDR-df0127` contradicts "cheaper decomposition"	E2
atlas `LEAD-ICB-nonlinear-frob-stable-bases`	does not cite `research/hamming_ideal_pdp_20260930` (same idea, H1 refuted, exponent 1.9–2.5 vs 0.8)	E3
atlas `LEAD-ICE-hybrid-guess-determine-l7`	"1/116 to 1/3,400" merges two exponent assumptions; "K unmeasured" is outdated (PR #2164 measured it: a column touch costs 4.5× a rho step)	E2
my LEAD-15 dossier	four corrections in §4.1	E5
cover census README	"Õ(q) is a heuristic, not a theorem" is outdated (Diem 2012)	E5

8. What I verified and what I did not

Verified by me: the phase shares and the m=3/m=2 ratios (recomputed from the committed record, agree with E3 to about one point); the raw base and h = 4 of the volcano specs and the existence of the projected plug-in; the Diem exponents (search result only, not the theorem text); the atlas counts (153 leads; 74 closed, 53 open_next_step, 17 open_blocked, 9 open_unexamined).

Relayed from evaluators without re-checking: every other number and quotation in §4, §5.1 and §7. Primary sources they read in full or in the relevant sections are listed on each card (E5: Diem JTNB 2012, Diem arXiv:math/0606607, Diem–Thomé 2008, Smith arXiv:0806.2995, Joux–Vitse, Tian arXiv:2012.07173; E4: the Bochum hashed-factor-base paper and the OpenAI preprint it depends on; E2: Gebregiyorgis' thesis, Faugère–Huot–Joux–Renault–Vitse, Petit–Kosters–Messeng). Abstract-only or inaccessible: Wagner 2002, ePrint 2026/384, ePrint 2025/783, Corrigan-Gibbs–Kogan 2018, Iijima–Momose–Chao 2015, Gaudry 2009, GGMP 2020, Minder–Sinclair, Bettale–Faugère–Perret.

Searched, nothing found: the shared session logs of 2026-10-24 to 2026-10-31 contain no mention of LEAD-19, `prime-projected-abscissa` or a volcano projected-base rerun, and no run of LEAD-15's harvest; they do contain the 2026-10-26 Hess audit above and a triage of an 8-strategy plan against the atlas (a different set of strategies; its decisions agree with the saturation seen here).

Not verifiable here: every atlas `evidence:` path in crypto-autoresearcher or added after 2026-10-24 (the local checkouts are stale); the ml-cryptanalysis `fb-koblitz-*` and `fb-prime-*` reports; whether OpenAI's Theorem D.1 is correct; whether Koblitz Table A/C bases are class-projected.

9. What to run next, by decisive value per cost

1. **Volcano rerun with `prime-projected-abscissa`** (≈ 2.4–7.7 CPU-h, no new code): tests whether LEAD-19's mechanism carries to the prime abscissa base and whether the volcano effect is real. If it carries, it is the next candidate improvement to a baseline, class engineering.
2. **LEAD-15 builder and harvest** (builder 2–3 person-days, run < 1 CPU-h): the only exponent test.
3. **Validation-gate artefacts for LEAD-19**, then the three-summand rows (already listed in its §8).
4. Everything else is a stage diagnostic or a null by derivation; run only when idle capacity exists.

None of these is a claim. Each becomes an improvement only if a measured, verified S beats the baseline named before the run, with the gap to matched rho reported as distance remaining.

10. Files

`README.md` (this), `report.pdf`, `figures/`, `phase_shares.{py,json,md}`, `plot_evaluation.py`, `candidate_judgements.tsv`, `atlas_match_log.tsv`, `cards/` (shared brief and the five unedited evaluator outputs), `evaluator_scripts/` (the exploratory scripts the evaluators wrote; one is a single-file Rust toy, labelled exploratory by its author), `atlas_drafts/`, `build.sh`.